INCH-POUND

MIL-A-70754 (AR)
10 April 1989

MILITARY SPECIFICATION

ADAPTERS, PALLET, METAL, FOR METAL CONTAINER FOR 105MM AND 120MM TANK AMMUNITION

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the fabrication of a number of the selected items, quality assurance provisions and packaging of the special adapters for tank ammunition pallets. The special adapters will consist of a top and a bottom that will unitize the 105mm and 120mm tank ammunition cartridge.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbooks.. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8140

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1261	-	Welding Procedures for Constructional Steel.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation

DRAWINGS

U.S. Army Defense Ammunition Center and School

AC200000423	-	Chemical Agent Resistant Coating (CARC) Finishing Requirements for Metal Pallets and/or Pallet Adapters
AC200000469	-	Top Assembly - Pallet Adapter, PA116 Container.
AC200000470	-	Bottom Assembly - Pallet Adapter, PA116 Container.
AC200000471	-	Top Assembly - Pallet Adapter, PA117 Container.
AC200000472	-	Bottom Assembly - Pallet Adapter, PA117 Container.
AC200000501	-	Adapter - Pallet, PA116 Container
AC200000502	-	Adapter - Pallet, PA117 Container

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2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-E8	-	Tension Testing of Metallic Materials, Standard Method of
ASTM-D3951	-	Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials. 1916 Race Street, Philadelphia, PA 19103).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Adapters. The adapter assemblies shall comply with all requirements specified on drawings (Dwgs) AC200000469, AC200000470, AC200000471 and AC200000472 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Welding. The welding shall comply with the requirements of MIL-STD-1261.

3.3.1 Tensile strength. The tensile strength of the weld shall equal or exceed the tensile strength of the material.

3.4 Painted surface. The painted surfaces of the adapters shall comply with the requirements of drawing AC200000423.

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3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship.

3.6.1 Adapters. The adapters shall be regular, smooth, and free from wrinkles, cracks, rough spots, burrs, sharp edges and any other defect that might affect the serviceability, durability, safety and appearance of the adapters.

3.6.2 Paint coating. The finish coat shall cover all required surfaces and be continuous with the possible exception of a few light scratches not exposing the base metal or the primer coat.

3.6.3 Parts. All parts shall be free of chips, dirt, grease, rust and foreign material. The cleaning method used shall not be injurious to any of the parts nor shall any of the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 6 each assemblies as per Dwgs. AC200000469, AC200000470, AC200000471, and AC200000472, as applicable.

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Adapters		SHEET 1 OF 1		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Top Assembly PA116 container/PA117 container, Dwg. AC200000469/AC200000471, Examination for defects.	6/6		3.2	4.4.2.1
	Bottom Assembly PA116 container/PA117 container, Dwg. AC200000470/AC200000472, Examination for defects.	6/6		3.2	4.4.2.2

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70754 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Top Assembly - Pallet Adapter, PA116 Container Top Assembly - Pallet Adapter, PA117 Container		AC200000469/AC200000471
			NEXT HIGHER ASSEMBLY AC200000501/AC200000502

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall width		1.0%	3.2	Gage
102	Weld tensile			3.3.1	4.5.1
103	Marking, missing, incorrect or illegible		1.0%	3.2	Visual
104	Distance between stacking lug (4 places)		1.0%	3.2	Gage
Minor					
201	Protective finish damaged		1.0%	3.2	Visual
202	Primer thickness		1.0%	3.2	Gage
203	Top coat thickness		1.0%	3.2	Gage
204	Weld missing, incomplete or incorrect		1.0%	3.3	Visual
205	Poor workmanship		1.0%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70754 (AR)

PARAGRAPH	TITLE				DRAWING NUMBER
4.4.2.2	Bottom Assembly - Pallet Adapter, PA116 Container Bottom Assembly - Pallet Adapter, PA117 Container			SHEET 1 OF 1	AC200000470/AC200000472
					NEXT HIGHER ASSEMBLY
					AC200000501/AC200000502
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall width		1.0%	3.2	Gage
102	Weld tensile			3.3.1	4.5.1
103	Marking, missing, incorrect or illegible		1.0%	3.2	Visual
Minor					
201	Protective finish damaged		1.0%	3.2	Visual
202	Primer thickness		1.0%	3.2	Gage
203	Top coat thickness		1.0%	3.2	Gage
204	Weld missing, incomplete or incorrect		1.0%	3.3	Visual
205	Poor workmanship		1.0%	3.6	Visual

NOTES:

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4.4.3 Testing.

4.4.3.1 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.5 Methods of inspection.

4.5.1 Tensile strength. The tensile strength shall be determined in accordance with ASTM-E8. One assembly shall be randomly selected from the lot. Three specimens of the lowest gage material thickness shall be selected from the assembly for determination of the material tensile strength. Also, three specimens shall be randomly selected at right angles to the weld and the tensile strength of the weld shall be determined. Each of these values shall be compared with the lowest determined value of the material tensile strength for compliance with requirement of 3.3.1. If any of the samples fails to meet the requirement, the lot shall be rejected.

5. PACKAGING

5.1 Preservation and packaging. None required.

5.2 Packing.

5.2.1 Commercial. The adapter shall be packed in accordance with ASTM-D-3951.

5.3 Marking.

5.3.1 Commercial. Shipping marking shall comply with ASTM-D3951-82 and the following additional instructions. No shipping marking shall be placed directly on the adapters. Commercial shipping tags attached with wire shall be used unless otherwise specified. The following information shall be shown: Item name and model number, part number and revision, contract number, and date, name and address of manufacturer. If adapters are bundled, boxed or bulk-contained, the gross weight and quantity shall be included. The above information may be placed on bulk-containers or boxes by labeling or stenciling.

5.4 Shipping. When adapters from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing lots in transit.

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6. NOTES

6.1 Intended use. These adapters are intended for use with the standard pallet and with the metal tank ammunition container for the 105mm and 120mm tank ammunition cartridge.

6.2 Acquisition requirements. see MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Approval of equivalent test methods. Prior approval of the contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-A (D), Picatinny Arsenal, NJ 07806-5000. This description should include but not be limited to the accuracy and precision and drawings of any special equipment required.

6.5 Subject term (key word) listing.

Ammunition
Cartridge
Container
Metal
Pallet
Tank
Adapters

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 8140-A759)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-70754 (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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